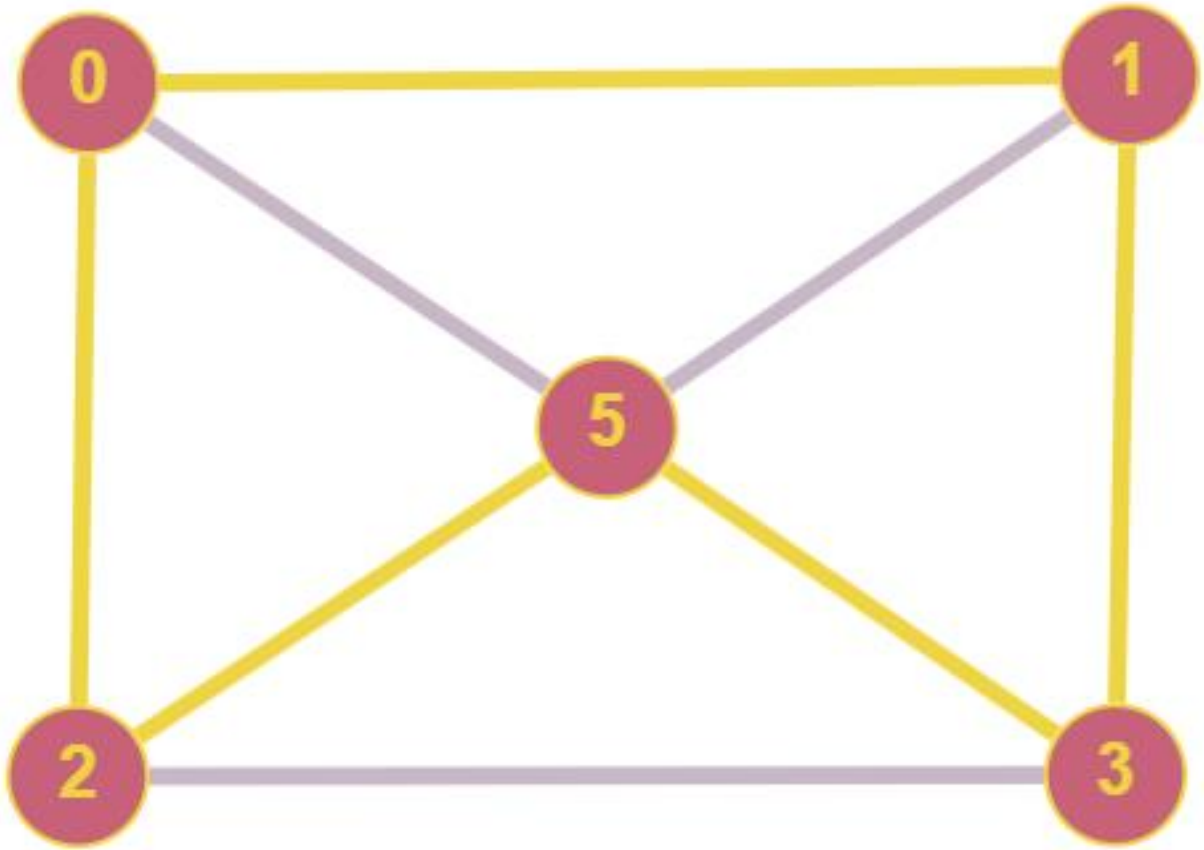
Hamiltonian Cycle Problem

Jurij Cerar

TU Wien, Austria

Problem Description

- Visit each vertex exactly once, ending on the same vertex as we began.
- Encode problem and feed to SAT solver
- Output if graph is Hamiltonian and if so write out the cycle



Tools Used

- Python v3.11.9
- PySAT
- **Efficient SAT Techniques for Relative Encoding of Permutations with Constraints** by Miroslav N. Velev and Ping Gao
- FHCP Challenge Set

Implementation

- HCP format files (dimension & edges)
- S_{ij} - vertex i is the direct predecesor of j (successor/edges)
- O_{ij} - vertex i is the predecesor of j (variable order)
- Each has 4 properties that need to be encoded
- Relationship constraint $S_{ij} \Rightarrow O_{ij}$

Problems

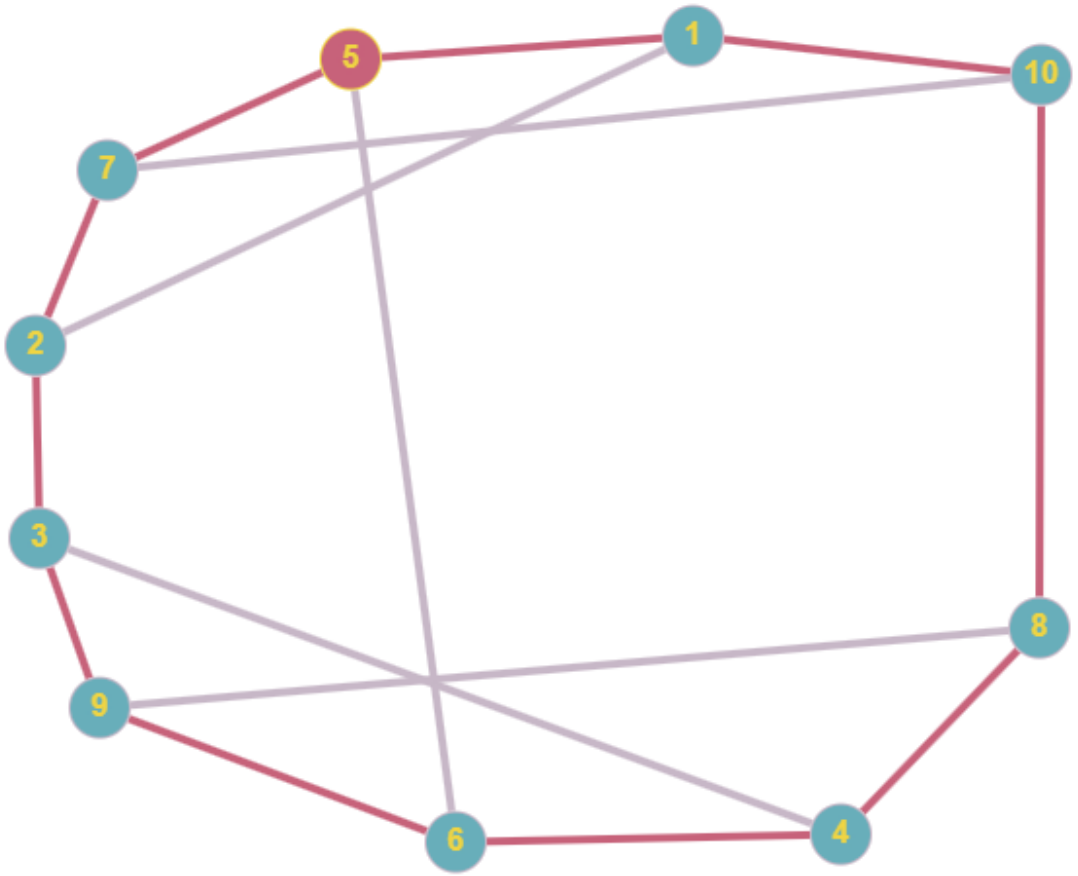
- pySAT dependencies didn‘t work (later fixed)
- Mutual exclusion constraint for successor variables added
- First vertex should not have relationship constraint in order to complete the cycle
- Hard to debug on large graphs

Example

- 210 vertice graph took 17s

```
COMMENT : Example graph with 10 vertices.
TYPE : HCP
DIMENSION : 10
EDGE_DATA_FORMAT : EDGE_LIST
EDGE_DATA_SECTION
1 2
1 5
1 10
2 3
2 7
3 4
3 9
4 6
4 8
5 6
5 7
6 9
7 10
8 9
8 10
-1
EOF
```

```
Is solution valid hamilton graph?  True
SOLUTION:  [1, 10, 8, 4, 6, 9, 3, 2, 7, 5, 1]
```



END

THANK YOU!